

Measure of central tendency

28-2-24

ARITHMETIC MEAN

- The arithmetic mean is the simplest measure of central tendency.
- It is simply called as Mean.
- It can be defined as the sum of all the observations divided by the number of observations.
- It can be obtained by the formula:

$$\bar{X} = \sum_{i=1}^n X_i / n$$

Where;

X_i = observation;

n = no. of observation

ARITHMETIC MEAN

- For frequency table (for discrete data as well as for class interval table) the mean is calculated by using formula:

$$\bar{X} = \frac{\sum_{i=1}^n f_i X_i}{\sum_{i=1}^n f_i}$$

Where X_i = value of the observation or class mark

f_i = frequency for i^{th} observation or class interval

MEDIAN

- Median is another measure of central tendency.
- It can be calculated for quantitative data as well as qualitative data.
- It is given by the value of the middle observation after arranging them into ascending or descending order.
- For the data in raw format the median is obtained by using the formula:
Median (M_d) = Value of $\{(n+1)/2\}^{\text{th}}$ term; if n is odd.
= [Value of $(n/2)^{\text{th}}$ term + Value of $\{(n/2)+1\}^{\text{th}}$ term]/2; if n is even.

- For discrete data, Median is the observation having cumulative frequency just greater than $(N/2)$.
- For class interval data, median is obtained by:

$$Md = l_0 + \left\{ \frac{(N/2) - C}{f_0} \right\} h$$

l_0 = Lower limit of median class, f_0 = frequency of median class, C = cumulative frequency before median class, h = width of median class

MODE

- ✖ It is another measure of central tendency.
- ✖ It also can be calculated for the qualitative as well as quantitative data.
- ✖ It is the value of term having Highest frequency in the data i.e. the value of term which occur most in the data.
- ✖ For Raw (or discrete) data it is the value of observation having highest frequency.
- ✖ For class interval data it is computed with the help of formula:

$$Mo = l_0 + \left(\frac{f_0 - f_{-1}}{2f_0 - f_{-1} - f_1} \right) h$$

- ✖ Where, f_{-1} = frequency before mode class, f_1 = frequency after mode class, f_0 = frequency of mode class, h = width of mode class, l_0 = lower limit of mode class

- ✖ The following are the arterial activates partial thromboplastin time for 35 patients.

20.7	29.3	34.4	56.6	22.5	29.7	24.2
31.2	38.3	28.5	22.8	44.8	41.6	46.1
24.9	29.0	30.1	33.9	39.7	45.3	33.6
22.9	20.3	28.4	35.5	22.8	54.7	22.5
52.4	20.9	46.1	35.0	46.1	22.1	50.1

EXAMPLE (MEAN)

The sum of observations is:

$$\sum_{i=1}^N X_i = 1187$$

$$N=35$$

So,

$$\overline{X} = 1187 / 35 = 33.91$$

After arranging the data into ascending order we get:

20.30	22.80	29.30	35.00	46.10
20.70	22.90	29.70	35.50	46.10
20.90	24.20	30.10	38.30	46.10
22.10	24.90	31.20	39.70	50.10
22.50	28.40	33.60	41.60	52.40
22.50	28.50	33.90	44.80	54.70
22.80	29.00	34.40	45.30	56.60

✖ No. of observation = 35; which is a odd number.

So,

$$\begin{aligned} Md &= \text{Value of } \{(35+1)/2\}^{\text{th}} \text{ term} \\ &= \text{Value of } 18^{\text{th}} \text{ term} \\ &= 31.20 \end{aligned}$$

Following data represents the wages of the workers in a factory. Compute the different measures of central tendency.

Wages	No. of workers
170 – 180	52
180 – 190	68
190 – 200	85
200 – 210	92
210 – 220	95
220 – 230	100
230 – 240	70
240 – 250	28

Computation of Mean:

For computing the mean first class marks (X_i) are computed using the formula $X_i = (Upper\ Limit + Lower\ Limit)/2$

Then following table is constructed:

Wages	No. of workers (f_i)	Class Mark (X_i)	$f_i X_i$
170 – 180	52	175	9100
180 – 190	68	185	12580
190 – 200	85	195	16575
200 – 210	92	205	18860
210 – 220	95	215	20425
220 – 230	100	225	22500
230 – 240	70	235	16450
240 – 250	28	245	6860
Total	590		123350

Now the mean of the given data would be

$$\bar{X} = 123350/590 = 209.07$$

Computation of Mean:

Wages	No. of workers	X_i	d_i	$fidi$
170 – 180	52	175	-3	-156
180 – 190	68	185	-2	-136
190 – 200	85	195	-1	-85
200 – 210	92	205	0	0
210 – 220	95	215	1	95
220 – 230	100	225	2	200
230 – 240	70	235	3	210
240 – 250	28	245	4	112
Total	590			240

Then the Mean of the data will be:

$$\bar{X} = 205 + (240/590) \cdot 10 = 205 + 4.07 = 209.07$$

Computation of Median:

For computing the median first the median class is decided. It is the class where the less than type cumulative frequency become just greater than $N/2$. In the example consider the $N/2 = 295$.

Wages	No. of workers	Cumulative Frequency
170 – 180	52	52
180 – 190	68	120
190 – 200	85	205
200 – 210	92	297
210 – 220	95	392
220 – 230	100	492
230 – 240	70	562
240 – 250	28	590

Median Class

Computation of Median:

As for the class interval 200-210 the cumulative frequency (297) is just greater than 295, The median class will be 200-210. Therefore,

$$l_0 = 200, f_0 = 92, C = 205, h = 10$$

$$\begin{aligned} Md &= 200 + \{(295 - 205) \cdot 10 / 92\} \\ &= 200 + 9.78 \\ &= 209.78 \end{aligned}$$

For computing the mode first the mode class is to be decided. It is the class where the maximum frequency occur.

Wages	No. of workers	Cumulative Frequency
170 – 180	52	52
180 – 190	68	120
190 – 200	85	205
200 – 210	92	297
210 – 220	95	392
220 – 230	100	492
230 – 240	70	562
240 – 250	28	590

Mode
Class

In the example consider maximum frequency is 100 for the class interval 220–230. Therefore mode class will be 220–230 and:

$$l_0 = 220, f_0 = 100, f_{-1} = 95, f_1 = 70, h = 10$$

$$\begin{aligned}
 Mo &= 220 + \{(100-95)/(200-95-70)\} * 10 \\
 &= 220 + \{5 * 10 / 35\} \\
 &= 220 + 1.43 \\
 &= 221.43
 \end{aligned}$$

